

A Few Facts about **ARP** Spoofing

Wireless sniffing involves the capture of all the packets that pass through a given wireless network interface. Sniffing is one of the most effective techniques used when attacking a network. Active sniffing involves injecting packets into the network, which leads to traffic. This article deals with active sniffing and ARP spoofing.



In a local area network or LAN, the term sniffing refers to capturing packets transmitted over the network. An attacker may use either the passive or active method of sniffing. Passive sniffing is done on a HUB environment, using promiscuous mode.

ARP spoofing is used for active sniffing. But before discussing ARP spoofing, let's be clear about the ARP protocol.

Address resolution protocol (ARP)

ARP refers to the process of finding the MAC address of a computer in a network. There are two types of ARP messages that can be sent:

- ARP request
- ARP reply

In an ARP request message, the sender broadcasts the ARP request to all computers in a subnet to know the MAC address of the desired IP address.

In an ARP reply, the target system sends the reply to the sender, giving the MAC address. The ARP reply is a unicast response.

An ARP cache

To reduce the number of ARP requests, the system maintains a finite ARP cache for a short period of time. This contains the IP-to-MAC address mapping. When one system wants to send a message to another, it first checks its ARP cache. You can view the ARP cache of your computer using the following command:

```
arp -a
```

ARP spoofing

ARP spoofing, also known as ARP cache poisoning, is a type of attack in which the MAC address of the victim machine in the ARP cache of the gateway, along with the MAC address